

MNL-Bandit

Setting:

- Set of N substitutable products with known prices r_i and unknown time varying attraction parameters ω_i^t .
- At each time round t , the decision maker chooses an assortment $S \subset [N]$ of at most K products then product i is purchased w.p. $\omega_i^t / (1 + \sum_{j \in S} \omega_j^t)$ or no product is purchased w.p. $1 / (1 + \sum_{j \in S} \omega_j^t)$.
- We wish to maximize the total collected profit (Exploration-Exploitation trade off).

Applications: Online retailing, online advertisement etc.

Non-Stationarity Model

We consider adversarially changing attraction parameters ω^t .

Measures of Non-Stationarity: We consider two measures of non-stationarity.

- Number of switches:

$$L = 1 + \sum_{t=1}^{T-1} 1_{\{\omega^t \neq \omega^{t+1}\}}.$$

- Variation:

$$\Delta_{\infty}^K = \sum_{t=1}^{T-1} \|\omega^t - \omega^{t+1}\|_{L_{\infty}^{2K}},$$

where $\|x\|_{L_{\infty}^{2K}} = \sup_{S \subset [N]: |S| \leq 2K} \sum_{r \in S} |x_r|$ for $x \in \mathbb{R}^N$

Goal: Minimize the dynamic regret,

$$\text{DReg} = \sum_{t=1}^T \max_{S: |S| \leq K} \text{Rev}(S, \omega^t) - r^t.$$

where r^t is the revenue collected at time t .

Related Work

Stationary MNL-Bandit.

- Rusmevichientong et al. (2010) [9] and Saure and Zeevi (2013) [10]: $O(N^2 \log^2(T))$ and $O(N \log T)$ parameter dependent explore-then-commit algorithms respectively.
- Agrawal et al (2016, 2017) [1, 2]: $O(\sqrt{NT})$ UCB and Thomson sampling based algorithms respectively.
- Chen and Wang (2017) [6]: $\Omega(\sqrt{NT})$ lower-bound.

Non-Stationary Bandits.

Adversarial non-stationarity also considered for,

- Besbes et al. 2014 [4], Auer et al. 2019 [3]: Multi-armed bandits.
- Chen et al. 2021 [5]: Combinatorial semi-bandits.
- Chen et al. 2019 [7]: Contextual bandits.
- Cheung et al., 2018 [8]: Linear bandits.

Main Contributions.

- Upper-Bound.** We give an online algorithm achieving the dynamic regret upper-bound,

$$\tilde{O} \left(\min \left\{ \sqrt{NTL}, N^{\frac{1}{3}} (\Delta_{\infty}^K)^{\frac{1}{3}} T^{\frac{2}{3}} + \sqrt{NT} \right\} \right).$$

- Lower-Bound.** We show that our algorithm is optimal (up to a logarithmic factor) by giving matching lower-bound.

Algorithm

Intuitively, we run a stationary algorithm and periodically check for non-stationarity, when non-stationarity is detected, the algorithm restarts.

Stationary Algorithm.

We use a variant of the UCB algorithm of Agrawal et al. (2016) [1], which goes as follows,

- keep a UCB $\hat{\omega}_i$ for the attraction parameters of each product i .
- Offer optimal assortment w.r.t the UCBs until no-purchase happens (we say that an epoch ended).
- Update the UCBs at the end of the epoch using the number of purchases of product i as an estimator of ω_i .

Non Stationarity Tests.

We use the algorithm META of Wei and Luo (2021) [11]. This algorithm runs multiple copies of the stationary algorithm in a multi-scale manner and periodically tests for non-stationarity. In particular, it compares the total collected revenue to the UCB of the revenue (when ω_i is replaced by the UCB $\hat{\omega}_i$) and restarts whenever these diverge a lot.

Main Issue.

Bounding the dynamic regret boils down to bounding the regret collected by the stationary algorithm between two restarts.

Analysis

- In the stationary case, the number of purchases estimator is unbiased, geometric, and independent across epochs.
- Because ω^t , varies within an epoch, the estimator is now **biased**, has distribution that **depends on adversarial changes** and is **not independent** across epochs.

Our Approach:

- We bound the estimator between two “nice” random variables in the usual stochastic order.
- We derive concentration bounds for these “nice” random variables instead.
- We use these bounds to show that our original estimator is well concentrated around ω^t of any particular time step t in the corresponding epoch.
- Then use a special Lipschitz property of the revenue function to get our bounds.

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